

EXERCISE 3

CRYSTAL DIODE (SILICON –

GERMAN)

THEORETICAL PART

The operation of crystal diodes or simply diodes as is customary, but also of crystal triodes or transistors is based on the properties of crystalline solids called semiconductors. Semiconductors are bodies that

occupy a position between conductors and insulators in conductivity. The resistance of semiconductors depends very much on temperature and the temperature coefficient is negative, unlike metals.

This means that as the temperature increases, the resistance decreases and vice versa. The semiconductors used to make diodes and transistors are usually Silicon (Si) and Germanium (Ge).

If we add atoms of the element Arsenic (As) which is 5valent as an impurity to the atoms of a part of a pure semiconductor, e.g. Ge which is 4valent, then the 4 electrons of (As) will form covalent bonds with 4 valence electrons of 4 other atoms. Ge, then we will have an excess of one electron for each atom (As). Thus, a pure Ge semiconductor, in addition to the holes (electron vacancies, covalent bond sites) and the electrons, also acquired an excess number of electrons proportional to the impurities of the (As) atoms. Similarly, if we add atoms of the element Indium (In) to the atoms of the other part of the pure (Ge) as an impurity, then 4 covalent bonds will be formed again, only this time one bond will be incomplete due to the one less electron valence of (In).

This vacant position is a hole, which will be filled by a free electron and a hole will be created elsewhere. Thus, the (In) atoms that complete their covalent bonds with electrons are called acceptors, while the (As) atoms that give electrons are called donors.

The impurity (As) in pure (Ge) makes a new semiconductor with an excess of electrons that we call (N-type), while the impurity (In) in (Ge) makes the semiconductor have an excess of holes and we call it (P-type). (Fig.1) The holes and electrons are considered carriers of semiconductors equal in absolute value and opposite in polarity charges and the name of the types of semiconductor came from this polarity

P: Positive = positive, N: Negative = Negative.

Thus, in the semiconductor, the P – N junction was formed in which the transition from one

type of semiconductor to the other occurs gradually and not abruptly. For a better study of the phenomenon we will consider that the transition occurs abruptly at a separating flat surface (Fig. 2)

FIGURE 1 Flat symbolic representation of the structure of Ge atoms with N-type (As) and P-type (In) impurities

The properties of the P – N junction are essential for the operation of the crystal diode. A potential barrier, as it is called, is formed at the junction near the interface, because due to the mutual neutralization (reconnection) of the excess holes in the P-type semiconductor by the neighboring excess electrons of the N-type and vice versa, a region almost completely stripped of carriers (holes and electrons) was created, which we call the depletion region.

FIGURE 2 Representation of a P – N contact

The stripping region contains only immobile positive and negative ions but also a few carriers (holes and electrons) due to the breakdown of the bonds due to the effect of the ambient temperature on the contact. If we apply a constant voltage V as in (fig.3) (+) to P and (-) to N, then holes are pushed towards P with the result that the stripping region becomes much narrower and the carriers overcome the potential barrier, passing from one region to the other to create a current (in the right direction) of several (mA).

FIGURE 3

This connection of the contact is called forward bias. If we reverse the polarity of the source, (-) to P and (+) to N, then the opposite will happen and the stripping area will grow very large. The result of this bias is the almost zeroing of the current through the contact. A minimum (reverse) current of a few (mA) that will circulate in the circuit is due to the breakdown of some bonds of the pure semiconductor within the stripping area, due to the temperature of the contact. This connection is called reverse bias. The contact crystal diode is constructed in such a way that it presents mechanical resistance, is easily connected to the circuit and is not easily affected by temperature.

The diode as a component, its symbol and its correspondence with the P – N contact shown in figure 4.

FIGURE 4

The terminal connected to P is called the anode, while the terminal connected to N is called the cathode. Figure 5 shows two characteristics of Ge and Si diodes for the same applications.

FIGURE 5 Comparison of characteristics of typical Si and Ge diodes

It is observed that for a very small forward bias voltage they give a large current, while in reverse bias the current is very small. However, when the reverse bias increases significantly, it reaches the point V_{break} where the bonds of the pure semiconductor in the stripping region are broken down due to the creation of a strong electric field and the current increases sharply. If the diode has not been constructed to withstand this (such as Zeners), then it can be destroyed. **An ideal diode has the same characteristics as a**

switch. When forward biased it has zero resistance and zero voltage drop across its ends (switch in the ON state) and can pass current. When reverse biased it presents infinite resistance (OFF) and no current passes. However, an ideal diode does not exist. All diodes when forward biased present a voltage drop across their ends, when current passes through them and a slope in their characteristic after the voltage V_{break} , which means that they present some ohmic resistance. With reverse bias, some reverse current passes. The ohmic resistance presented by a diode can be divided into two types: the

static and dynamic resistance.

FIGURE 6

The static resistance R_{Dc} or R_{Dc} will be calculated for each point of the curve (after the voltage V_D) as the quotient of the voltage V_D corresponding to this point by the corresponding current I_D ,

$$R_{Dc} = \frac{V_D}{I_D} \text{ in (Ohm)}$$

a few tens (Ω) (static resistance in the forward direction). In the reverse direction the resistance is very large, i.e. a few (M Ω). The dynamic resistance

R_{Dn} or R_{Dn} will be calculated as the slope of the tangent (on the curve) at the point we want to calculate it or simply as the quotient of the voltage change ΔV_D at this point by the corresponding change in current

$$\Delta I_D$$

$$R_{Dn} = \frac{\Delta V_D}{\Delta I_D} \text{ in (Ohm)}$$

Note: the diode resistances are calculated on the straight part of the curve (after the point V_D) and approximately in its middle, usually at the operating point of Q. In this case, R_{Dc} is greater than R_{Dn} .

The diode is used in many electronic circuits such as rectifiers, charging, limiters, logic switches and others, for which if we want to understand their operation, we must know the operation of the diode, that is, when a diode conducts or does not conduct. A diode conducts as soon as we exceed the voltage V_D (theoretically as soon as we bias it forward) and does not conduct when we bias it reversely.

Summing up

In the following tables we summarize the behavior of the diode in forward and reverse directions. reverse polarization:

Right polarization

Since in reverse bias the resistance of the diode is very high, the voltage at its ends will result from the difference in the potentials of its ends, according to the overall circuit wiring.

LABORATORY PART

1. Purpose: The verification of the theory for contact diodes and the drawing of the characteristics I_f for forward and reverse polarization.
2. Instruments and devices:
 - a) DC power supply
 - b) Two voltmeters and an ammeter
3. Circuits:

4. a) We make the circuit of figure 7 and change the supply voltage according to table 1 below (correct diode polarization)

	Source(V)	0	0.1	0.3	0.6	0.9	1.5	2	4	7	9	12	15
Yes	VDF(V)												
	IDF(mA)												
Ge	VDF(V)												
	IDF(mA)												

Table 1

- b) We make the circuit of figure 8 and change the supply voltage according to the table below (reverse diode bias).

	Source(V)	0	1	2	4	6	8	10	12	14	15
Yes	VDR(V)										
	IDR(μ A)										
Ge	VDR(V)										
	IDR(μ A)										

Table 2

- c) Patterning of silicon (Si) diode characteristics by varying the supply voltage according to table 1 and table 2 for forward and reverse polarization

- d) Drawing the characteristics of the Germanium (Ge) diode in the same way as with the same tables 1 and 2 for forward and reverse polarization.

The engraving of the features should be designed to the same millimeter.

5. a) We carry out the connection of the following circuits, with D1 Silicon, and we take measurements on the voltmeters V1 , V2 , V3 , V4 and V5

FIGURE 9

V1=	Volts	V2=	Volts	V3=	Volts	V4=	Volts	V5=	Volts
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- b) Also explain theoretically the readings of the voltmeters V1 , V2 , V3 , V4 and V5 .

QUESTIONS

1. Explain the shape of the diode characteristics.
2. What is the difference between Germanium and Silicon diodes?
3. List the types and uses of diodes that you know.
4. Also from the V_{DF} and I_D values you filled in the previous table 1

(Silicon diode forward bias table **only**) calculate the values of R_{dynamic} and R_{static} completing the lower table 3. We note that the values $\Delta V(V)$ and $\Delta I(mA)$ are calculated by subtracting two consecutive values $V_{DF}(V)$ and $I_{DF}(mA)$ respectively from table 1

Source(V)	0.1	0.3	0.6	0.9	1.5	2	4	7	9	12	15
$\Delta V(V)$											
$\Delta I(mA)$											

Table 3

Then from the values in the above table draw the curves of R_{dynamic} and R_{static} in graph paper. The voltages and currents will must be in (Volt) and (mA). The curves that you will draw will have the following form about.

5. In the lower circuit, calculate the voltages of voltmeters V1 and V2 (D1 , D2 silicon real)

6. For the following circuits, evaluate the voltmeter readings. Obtain

Note that the diodes are all Si with $V_{\text{forward}}(\text{Si}) = 0.6 \text{ V}$

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EXERCISE^{OR}
4 ZENER DIODE
VOLTAGE STABILIZATION WITH ZENER DIODE

THEORETICAL PART

In the crystal diode exercise we saw that in forward bias the resistance of the diode is very low and thus allows a large current to pass through it. In reverse bias, the resistance of the diode is very large (infinite for an ideal diode) therefore, only a small current passes through it for a large reverse voltage.

However, at a certain reverse voltage V_z (Zener voltage), the current suddenly takes on large values, due to the breakdown of pure semiconductor bonds in the stripping region (Zener effect) and then chain breakdown of bonds (avalanche effect), due to the creation of a strong electric field in this region. However, the simple diode will be destroyed. Diodes that are constructed so that the high reverse

voltage and breakdown voltage (resulting in a large reverse current) do not destroy them are called Zener diodes. The Zener voltage (V_z) in each such diode is determined and depends on the density of impurities in the P – N junction. The higher this density, the lower V_z .

In forward bias, the Zener behaves like a common diode, as it does in reverse bias up to the voltage V_z . At the voltage V_z , however, the reverse current increases sharply, while the voltage changes little. This stability of the voltage, despite the change in current, makes the Zener very useful for stabilizing the voltage in limiting power supplies, etc.

Figure 1 shows the Zener diode characteristic curve as well as the its symbolism:

FIGURE 1

The change in voltage V_z min as V_z of $_{max}$ is so small that practically the voltage at the ends the Zener is considered constant, equal to V_z . The current does not change substantially up to the voltage V_z min where we have the minimum current $I_{z min}$ to have stabilization (in We will take our *calculations* $I_{z min} = 2mA$). The manufacturer gives the maximum current that

allowed to pass through the passage I_{zmax} and its limitation is done by a resistance that is connected externally in series with the diode.

The voltage V_z is usually given for the current I_z defined by the middle of straight line segment of the characteristic between the points corresponding to the I_{zmin} and I_{zmax} . The voltage V_z varies with temperature. In diodes of the same type the V_z may differ from diode to diode, with a deviation of 5 – 20%.

The maximum (allowable) power that can be consumed in the diode is given by

$$P_{zmax} = I_{zmax} V_z$$

Diodes with P_z are commercially available. I_{zmax} from ¼ W to 50 and for voltages

Zener stabilization V_z from 2 to 200 V. The dynamic

resistance R_z is defined by the slope of the reverse characteristic current at its operating point and is constant for its straight section

characteristic that is from the I_{zmin} to I_{zmax} : $R_z = \frac{V_z}{I_{zmax} - I_{zmin}}$

The main application of Zener diodes is voltage stabilization and used in DC power supplies to keep the voltage constant at the ends of the load, regardless of changes in the power supply voltage or changes in load and therefore its current.

Figure 2 shows a simple voltage stabilization circuit V_1 with a Zener, where the following relations apply:

$$I_R = I_z + I_L \quad (1)$$

$$V_1 = I_R R + V_z \quad (2)$$

$$V_z = I_L R \quad (3)$$

FIGURE 2

In this circuit we will have the following cases:

a) The voltage V_1 is constant and the current I varies due to a change in the load

R_L . The change in I_L can be equal to the difference I_{zmax} without going through

the Zener for maximum power P_{zmax} and I_{zmin} the minimum Zener current tolerable

stabilization, i.e. for constant V_z . Thus, relations (1) and (2) become:

$$\begin{aligned} I_{RMAX} - I_{LMIN} &= I \\ I_{RMIN} - I_{LMAX} &= I \end{aligned} \quad (1)$$

$$\begin{aligned} I_{LMIN} &= \frac{V_z}{R} - I_{RMAX} \\ I_{LMAX} &= \frac{V_z}{R} - I_{RMIN} \end{aligned} \quad (2)$$

b) The current I is constant and V_1 varies due to a change in the voltage of

network (220 V AC). An increase in V_1 causes an increase in the total current I and therefore increase in the Zener current I_z . Conversely, a decrease in V_1 causes a corresponding decrease in I_z .

In both cases the load current I_L remains practically constant. Thus the relations (1) and (2) become:

$$\frac{I_{RMAX} - I_{LMAX}}{I_{RMIN} - I_{LMIN}} = \frac{V_{1MAX} - V_{1MIN}}{V_{1MIN} - V_{1MIN}} \quad (1)$$

$$\frac{I_{RMAX} - I_{LMAX}}{I_{RMIN} - I_{LMIN}} = \frac{V_{1MAX} - V_{1MIN}}{V_{1MIN} - V_{1MIN}} \quad (2)$$

If Zener diodes are connected in series, they stabilize voltages that are multiples of the trend that can stabilize itself. If we want to stabilize the power to the stabilized voltage, we can connect similar Zeners in parallel. This connection is rarely used in practice, because it is difficult to find Zener completely similar so that the voltages at their ends do not show minor differences and the currents flowing through them to be equal.

Zeners can also be used a) for very high stabilization lower voltages than gas lamps, b) as common diodes and even for destabilization of lower voltages than those they can stabilize,

polarized in the right direction (stabilization at voltage V_Z), c) in the creation of voltages reference (voltage constant and independent of the operating conditions of the rest circuit with which we compare all the voltages that prevail in the circuit), d) as voltage limiter (clamp), e) where a constant thermal coefficient is required in various thermal stabilization circuits.

LABORATORY PART

Purpose: The verification of the theory for Zener diodes, the drawing of the characteristics for forward and reverse polarization and the experimental explanation of a voltage stabilization circuit with a Zener.

1. Make the circuit of figure 3. By varying the diode current
We measured the voltage at its ends.

Table 1

2. Make the circuit of figure 4. By varying the diode current
We measured the voltage at its ends.

Table 2

3. Plot the characteristic $IDF=F(VDF)$ for forward polarization based on the values in Table 1 and the characteristic $IDR=F(VDR)$ for reverse polarization based on the values in Table 2.
4. Make the circuit of figure 5 and by changing the input voltage V of the source, based on table 3 we measure the voltage at the ends of the Zener (and therefore at the load in which we have placed an intermediate value).

Table 3

5. In the same circuit (fig.5) vary the load current (by changing its value) and measure the voltage across the Zener based on table 4. (the V_Z at 15 V constant)

Table 4

6. What do you notice in the two previous tasks?

QUESTIONS

1. Explain the form of the Zener diode characteristic.
2. Analyze and explain the behavior of the circuit in Figure 5.
3. Where is the Zener diode used? Invent a circuit for each use (except stabilization).
4. In the circuit below we want V_{AB} to be at 10V and the current I to have values from 20 mA to 38 mA. Calculate the resistance R of the circuit with $V_{\text{supply}} = 20V$ and the power consumed by the Zener P_z is $\max$.
 valid $I_{ZMIN} = 2mA$ and $I_R = \frac{V_{ZMIN} - V_{AB}}{R}$ to I_{ZMAX}
 $I_R = \frac{V_{ZMAX} - V_{AB}}{R}$

FIGURE 2

5. From the reverse bias characteristic of the Zener that you drew in the assignment (par. 3), what is the dynamic resistance of the Zener? (calculate it graphically)